



Q1. A multimeter's resolution is defined in counts. How are these counts actually related to the resolution?

A1. The resolution of a multimeter is the smallest reading that can be shown on the display. In the past, the resolution of the multimeter used to be described in digits (like 4.5, 5.5). But these days the resolution is specified in counts. For example, let us look at 4.5-digit, or four-and-a-half-digit multimeter. The '4' represents full digits, meaning digits that can take up values from 0 to 9 on the display. The '0.5' represents the first digit of the display and the maximum it can read which is normally 0 to 1. Therefore 4.5 digits can show 00,000 to 19,999 digits, which make it a 20,000-count multimeter. To calculate the smallest reading that can be shown on the display, just divide the full-scale reading by the number of counts. For a 6000-count multimeter with maximum DC voltage range as 600V, the resolution will be $600/6000 = 0.1V$.

Q2. I am planning to implement a project based on RFID technology. Please help me in understanding how RFID technology works.

A2. There are a variety of RFID systems based on different operating principles, the most important being 'inductive coupling' and 'backscatter coupling.' These systems take advantage of the near-field and the far-field electromagnetic properties associated with an RF antenna, respectively. In both the cases, enough power can be transferred to the tag for its required operation.

Near-field RFID system. This type of system is sometimes also called inductive coupled system as it works on Faraday's principle of magnetic induction. The RFID reader produces

an alternating magnetic field in the vicinity by passing large alternating current through its coil. Now if a tag comes under the range of varying magnetic fields with its integrated smaller coil, an alternating voltage will be induced in the coil. This induced voltage is rectified and coupled to a capacitor. The capacitor gets charged and powers the semiconductor chip for its operation.

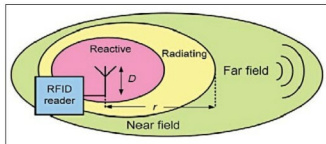


Fig. 1: Near field and far field

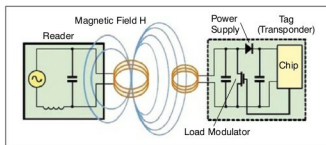


Fig. 2: Near-field RFID system

The tag sends back the data using load-modulation technique. As the voltage is induced by inductive coupling, if there is a load, the current will flow through the tag's coil, giving rise to its own small magnetic field. This small magnetic field will oppose the reader's field. Due to this change in field, the reader's coil will have a small amount of additional current flowing through it, which can easily be detected. The additional current in the reader's coil is actually proportional to load across the tag's coil, and hence the load modulation. This means that if the electronics in the tag applies a variable load to its own antenna, it can actually send an encoded signal representing the ID, which is detected at the RFID reader's side as

variation in the current through its coil. The modulation scheme can be chosen based on the number of bits to be transmitted, data rate and required redundancy.

Far-field RFID system. RFID tags based on far-field emissions capture electromagnetic waves propagating from the dipole antenna of the reader. Another smaller dipole antenna in the tag receives this energy as an alternating potential difference that appears across the arms of the dipole, which is further rectified by the diode and coupled to a capacitor. This stored energy in the capacitor powers the electronics. In this case, the tags are beyond the range of the near field and the ID cannot be transmitted back by load modulation. The technique used for communication in far field is called back scatter-

ing.

The antenna of the tag is designed precisely to tune it to a particular frequency so that it absorbs the maximum energy that reaches it at that frequency. Now if an impedance mismatch occurs at this frequency, the antenna of the tag will reflect back some energy. This reflected energy can be detected by sensitive radio receivers. The electronics of the tag varies the impedance over time to send back the encoded ID of the tag. This is generally done using a transistor across the dipole antenna and then tuning it partially on/off.

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